Comparison of Model to StreetLight Internal-Internal TAZ Trip Tables in Toledo Area

Presented to
The Ohio Travel Demand Model Users Group
September 8, 2017

By
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Methodology

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Project: Toledo-Test

Created by: greg.giaimo@dot.ohio.gov
Created on: 2017-07-28
Organization: Ohio DOT (ODOT) - Regional Subscription

Project Type: O-D Analysis (LBS Data)

Data Period: 2016:[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
Day Type:
1: Average Weekday (M-F)

Day Part:
0: All Day (12am-12am)


Metrics Version: R27-M35

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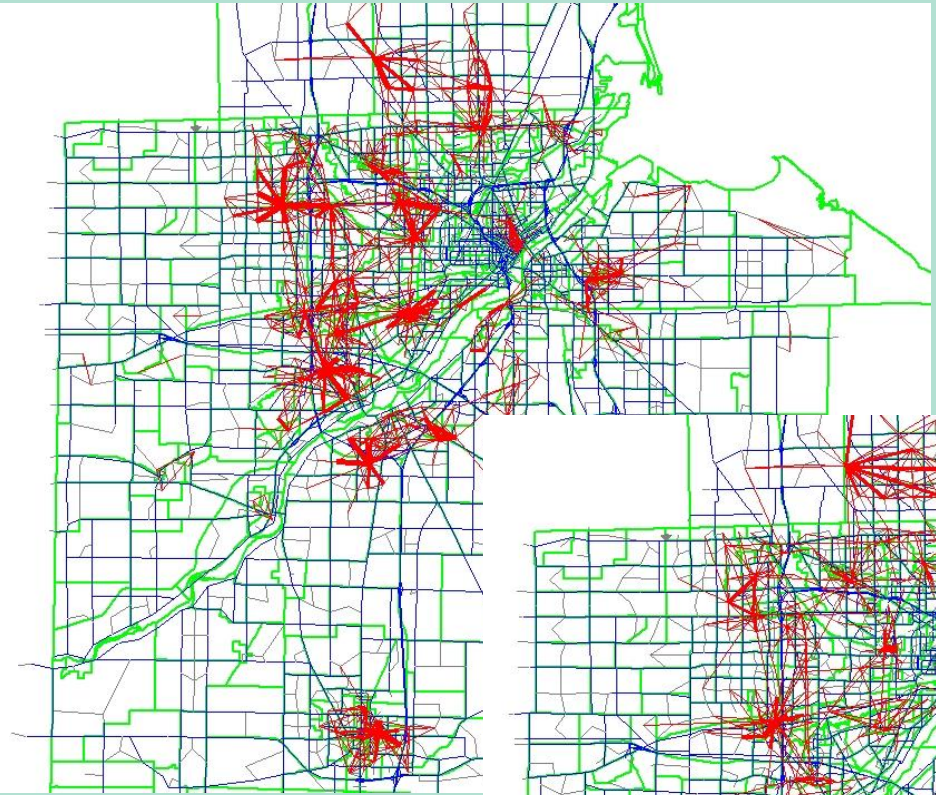
- Uploaded Shape File of TMACOG TAZ’s to StreetLight site
- Generated 3 Trip Tables
  1. Use “GPS Data” , Average Weekday Daily Trip Table for 2016
  2. Same but only Fall/Spring Months from Fall 2014 to Spring 2017
  3. Use “LBS Data”, Average Weekday Daily Trip Table for 2016
- Cube scripts import to Cube and factor to the total trips in the 2015 model trip table (only use the I-I portion of model)
- Make desire line plots on the network in Cube
- Another Cube script to calculate and report trip table coincidence ratios
- A final Cube script to factor the model TOD trip tables to the StreetLight Daily Totals for Assigning in OMS
- Assign and generate %RMSE and VMT stats

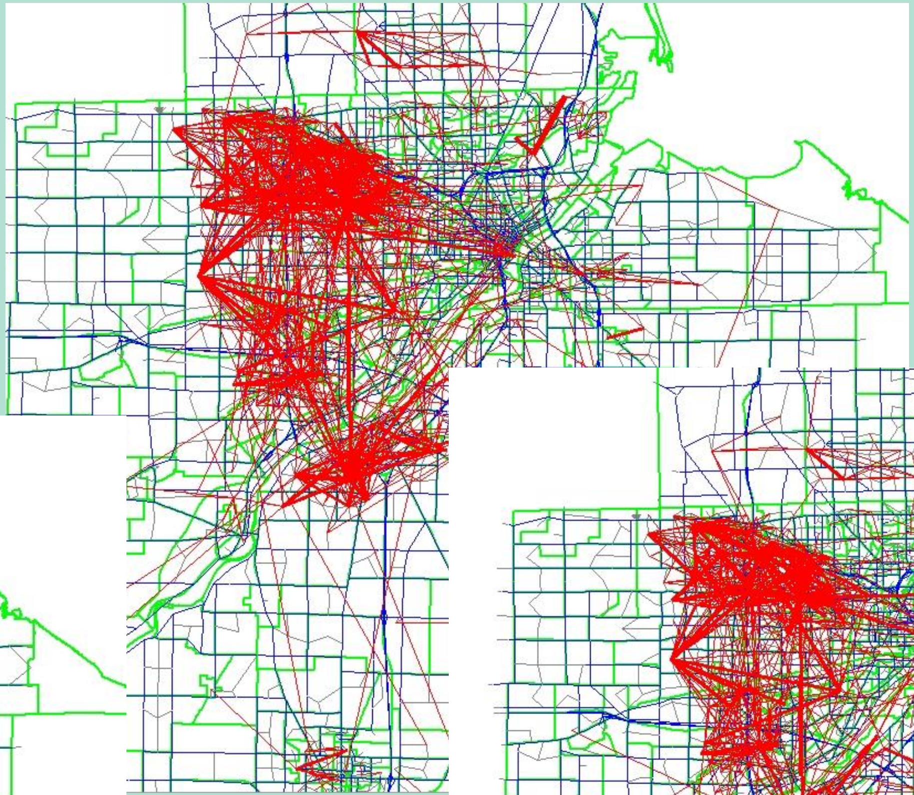
```
RUN PGM=MATRIX
FILEI MATI[1]=TOL15TBVEX.MAT
FILEI MATI[2]=GPS1417SF.MAT
  FILLMW MW[1]=MI.1.5 ;car (model)
  FILLMW MW[2]=MI.1.10 ;trk (model)
  FILLMW MW[3]=MI.2.1 ;car (raw streetlight)
  FILLMW MW[4]=MI.2.4 ;trk  (raw streetlight)
   IF(I<=920)
   JLOOP
    IF(J<=920)
     CTOTM=CTOTM+MW[1][J]
     TTOTM=TTOTM+MW[2][J]
     CTOTS=CTOTS+MW[3][J]
     TTOTS=TTOTS+MW[4][J]
    ENDIF
   ENDJLOOP
  ENDIF
  IF(I=920)
   CFACT=CTOTM/CTOTS
   TFACT=TTOTM/TTOTS
  ENDIF
  LOG PREFIX=FACTS,VAR=CFACT,TFACT
ENDRUN

RUN PGM=MATRIX
FILEI MATI[1]=GPS1417SF.MAT
FILEO MATO[1]=GPS1417SFFACT.MAT,MO=3,4,5,NAME=CAR,TRK,TOT
  FILLMW MW[1]=MI.1.1 ;car
  FILLMW MW[2]=MI.1.4 ;trk
  MW[3]=MW[1]*@FACTS.CFACT@
  MW[4]=MW[2]*@FACTS.TFACT@
  MW[5]=MW[3]+MW[4]
ENDRUN
```

Desire Lines at TAZ Level (Total Trips)

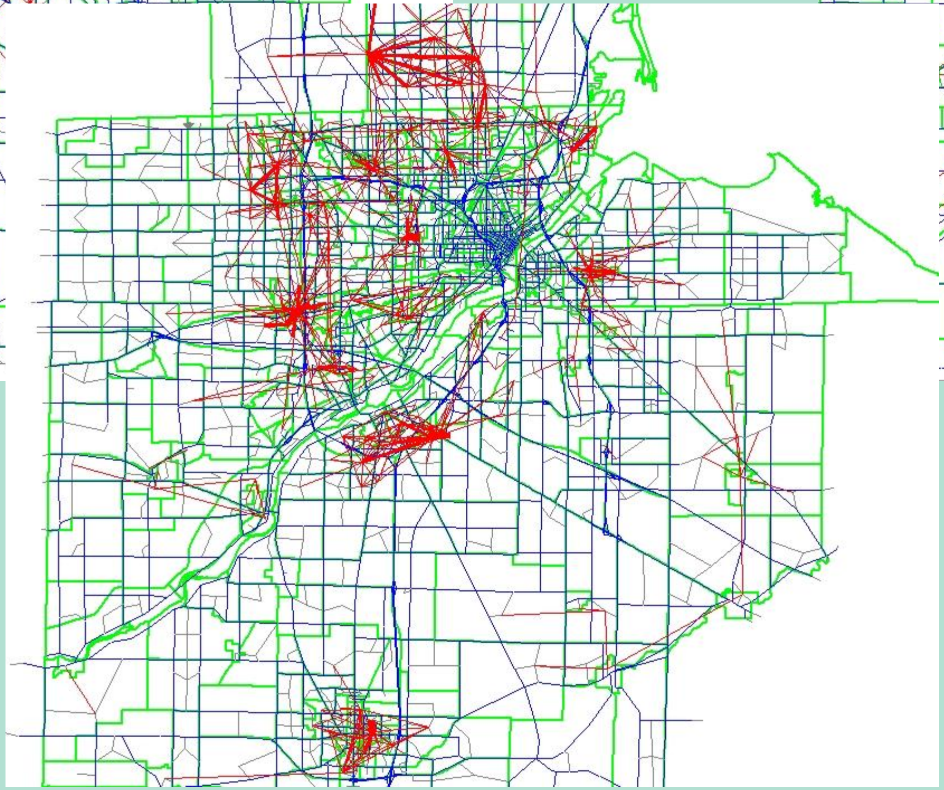
2015 Model



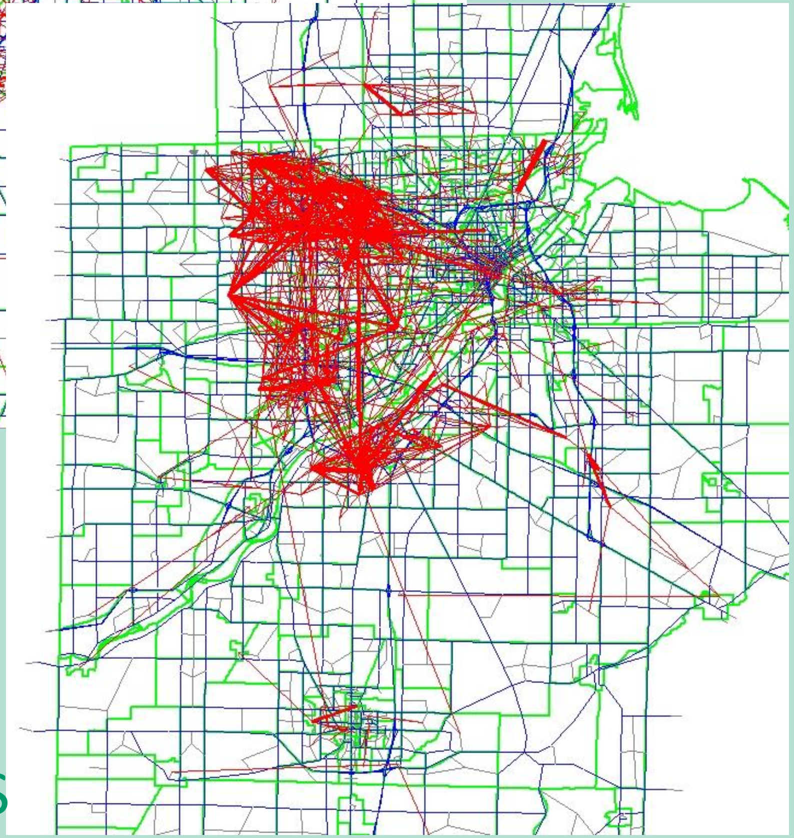
2016 GPS



2016 LBS

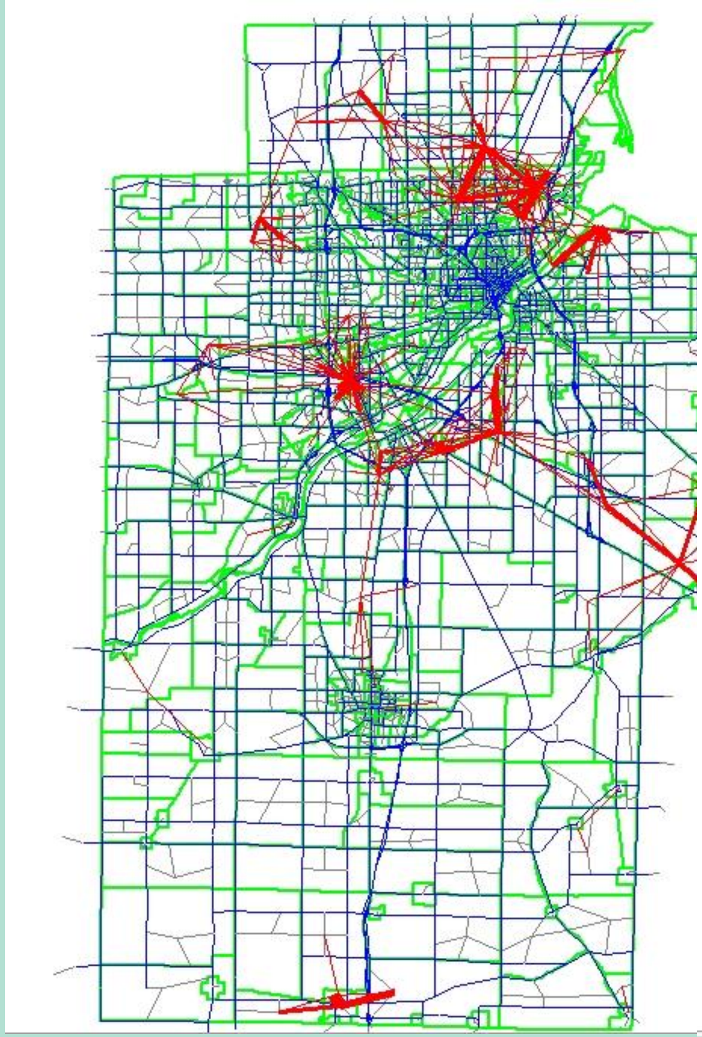


14-17 GPS

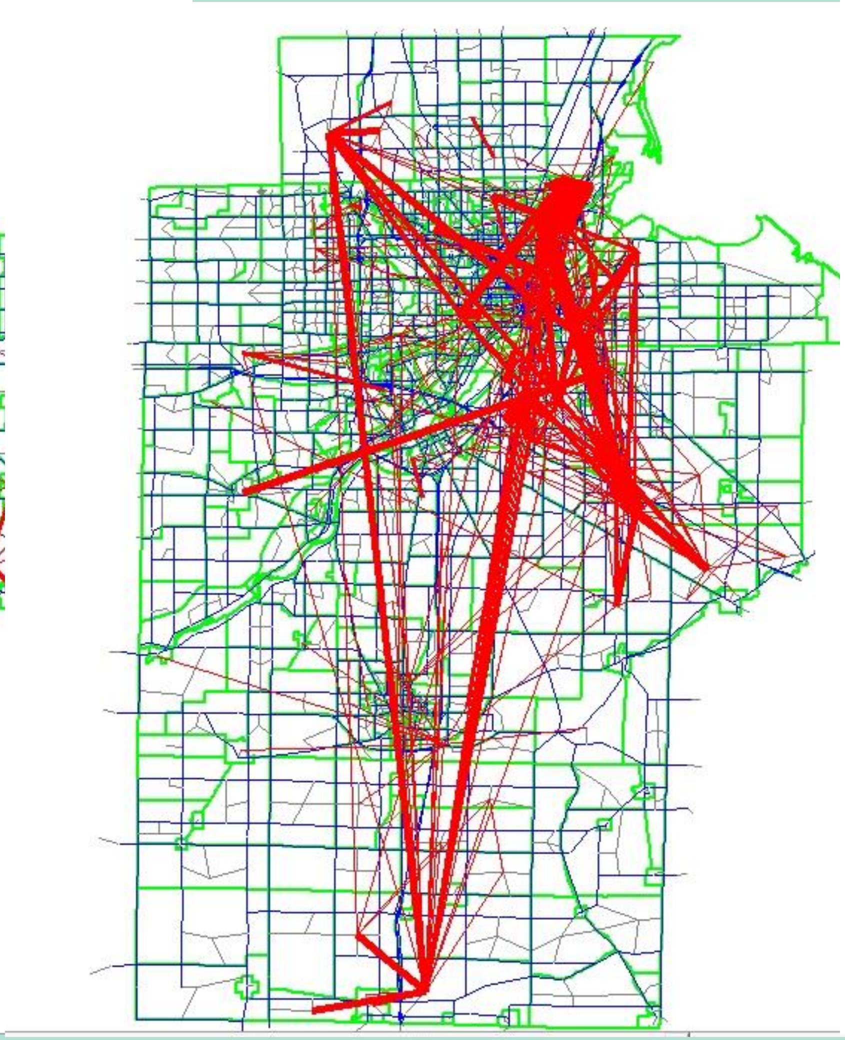


Desire Lines at TAZ Level (Truck Trips)

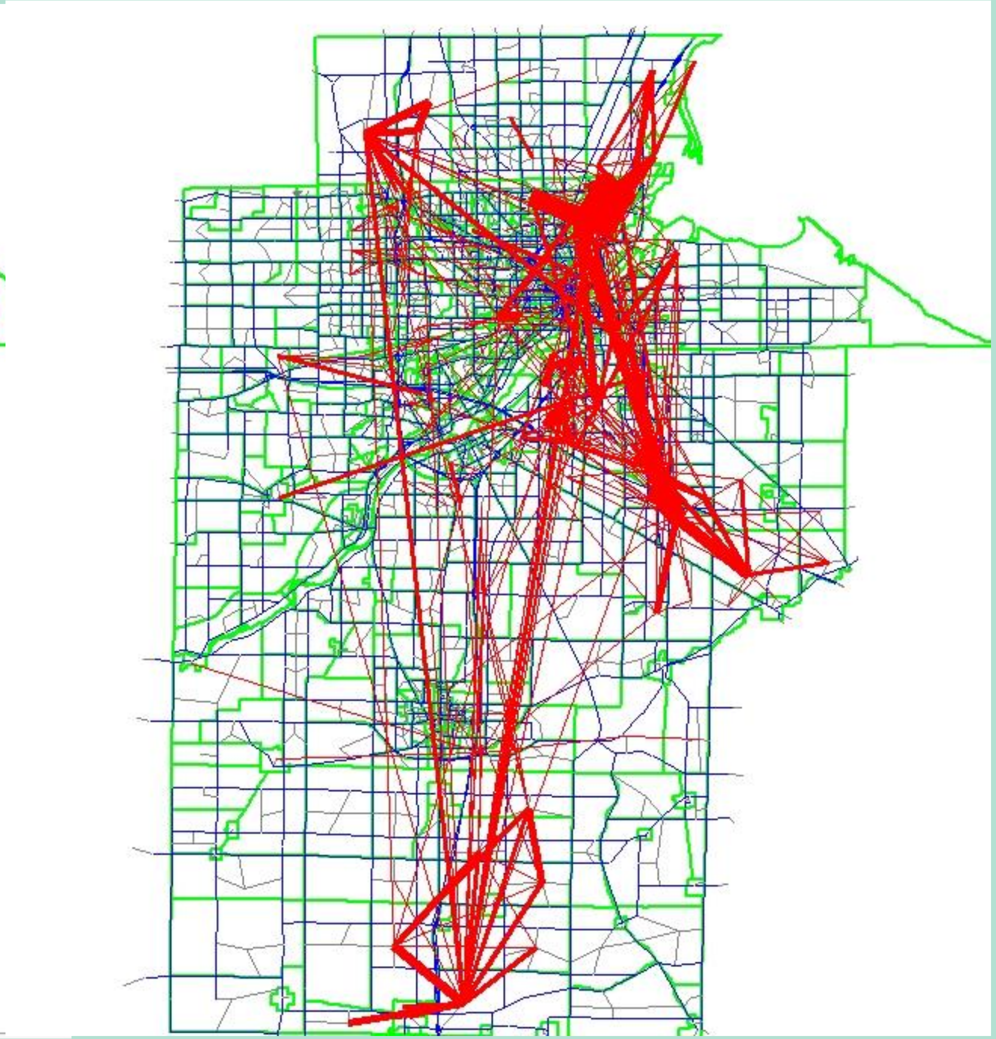
2015 Model



2016 GPS



14-17 GPS

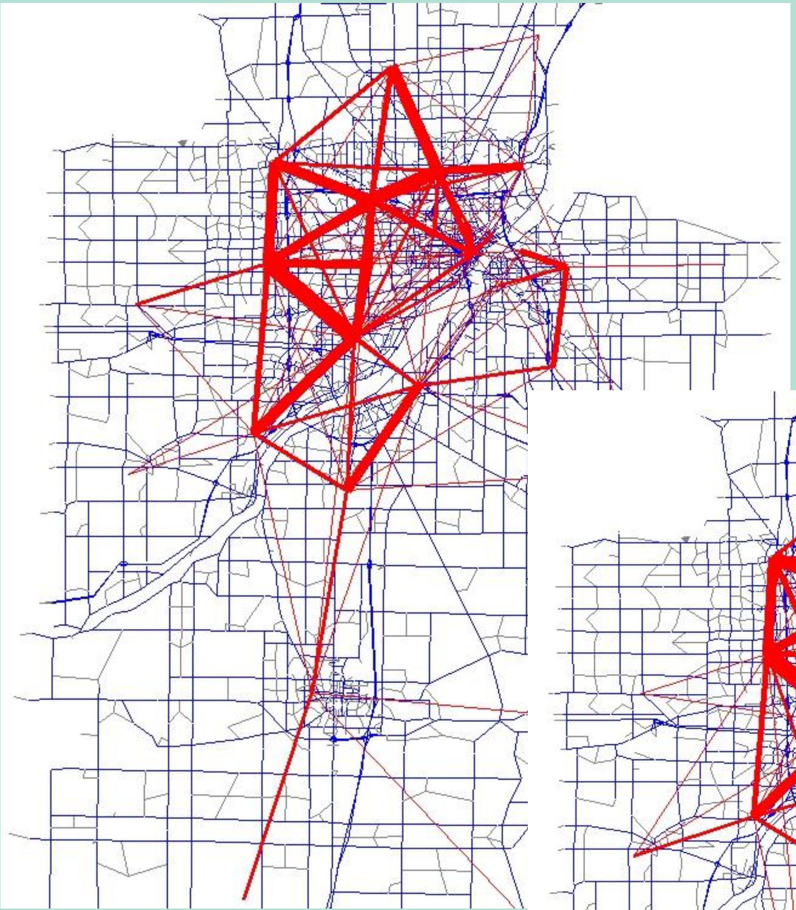


Desire Lines at TAZ Level – Some Thoughts

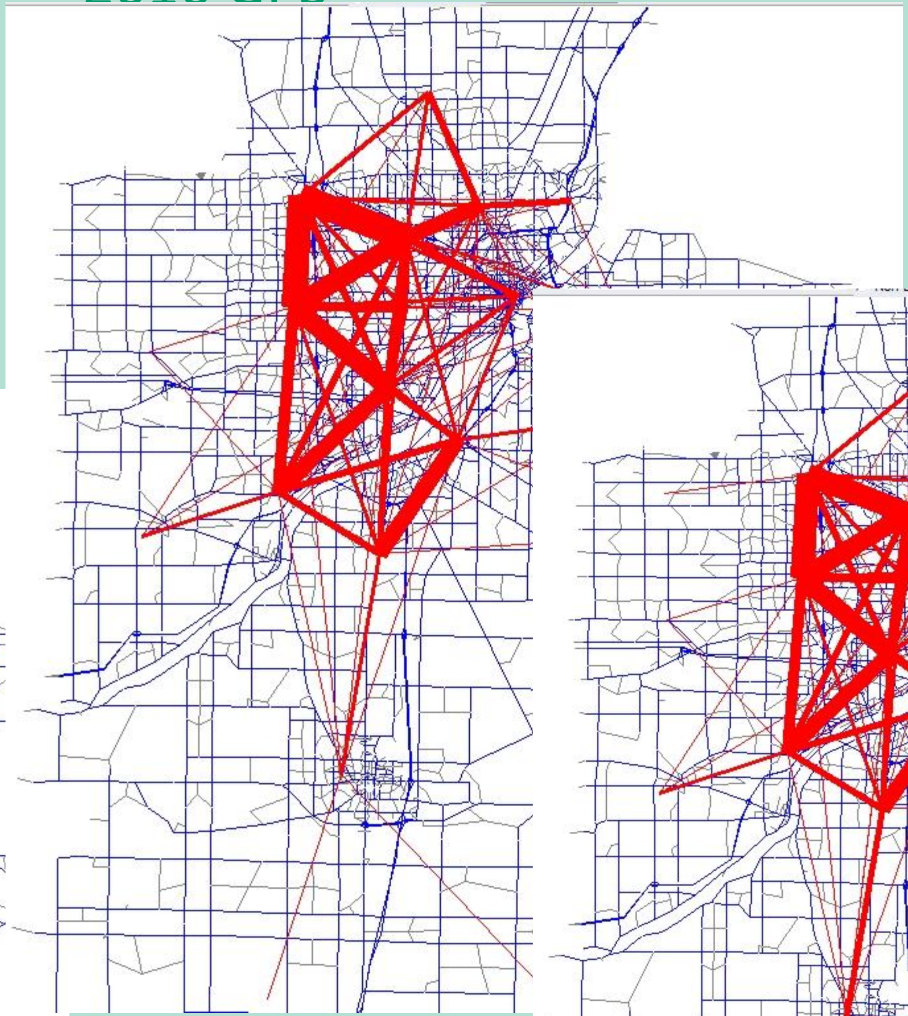
- They were all scaled to the same trip totals, so the apparent greater number of trips on the StreetLight maps is due to the sparseness of those tables compared to the model
- The model has many very low volume cells that don't show up on the plot, StreetLight has many zero cells but its populated cells (once scaled) tend to be higher
- The LBS data tends to match better (and StreetLight guidance suggests using that source for I-I trips)
- The GPS data appears to have some high income bias (people who own vehicles with onboard navigation systems) as there are many more trips in the rich areas (fortunately not a problem with trucks but would impact use for IE cars)
- Note, however, that you can't separate trucks in the LBS data, also the LBS data can't be used for pass through zones (i.e. external stations, which weren't analyzed here)
- Some of the long distance trip bias in the StreetLight data can be seen here, however, for trucks, the longer distance trips might be more reasonable than the model which for the MPO's at least isn't very good for I-I trucks
- On the other hand, the GPS truck data has many OD's at truck stops which the models don't (generally those are considered incidental in the models)
- The 2 GPS data sets seem reasonably comparable to one another

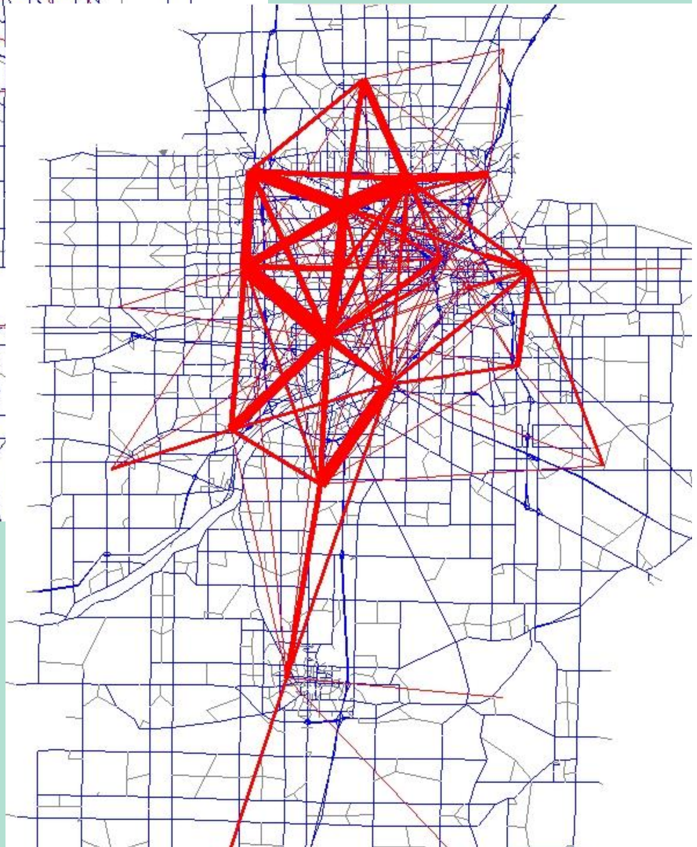
Desire Lines at District Level (Total Trips)

2015 Model

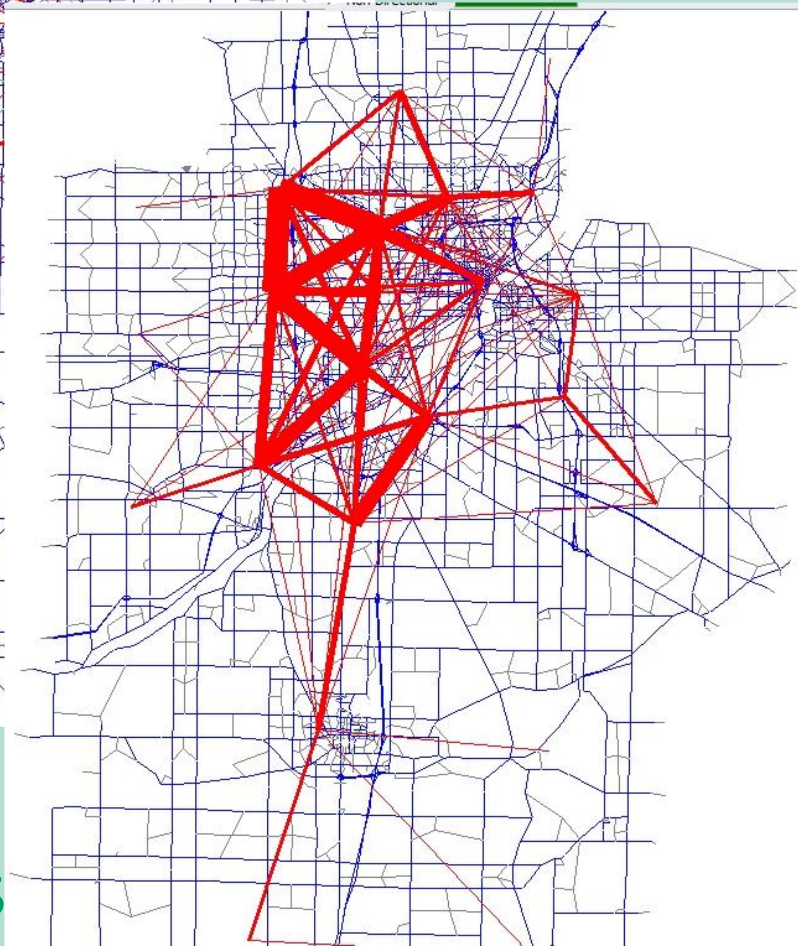


2016 GPS





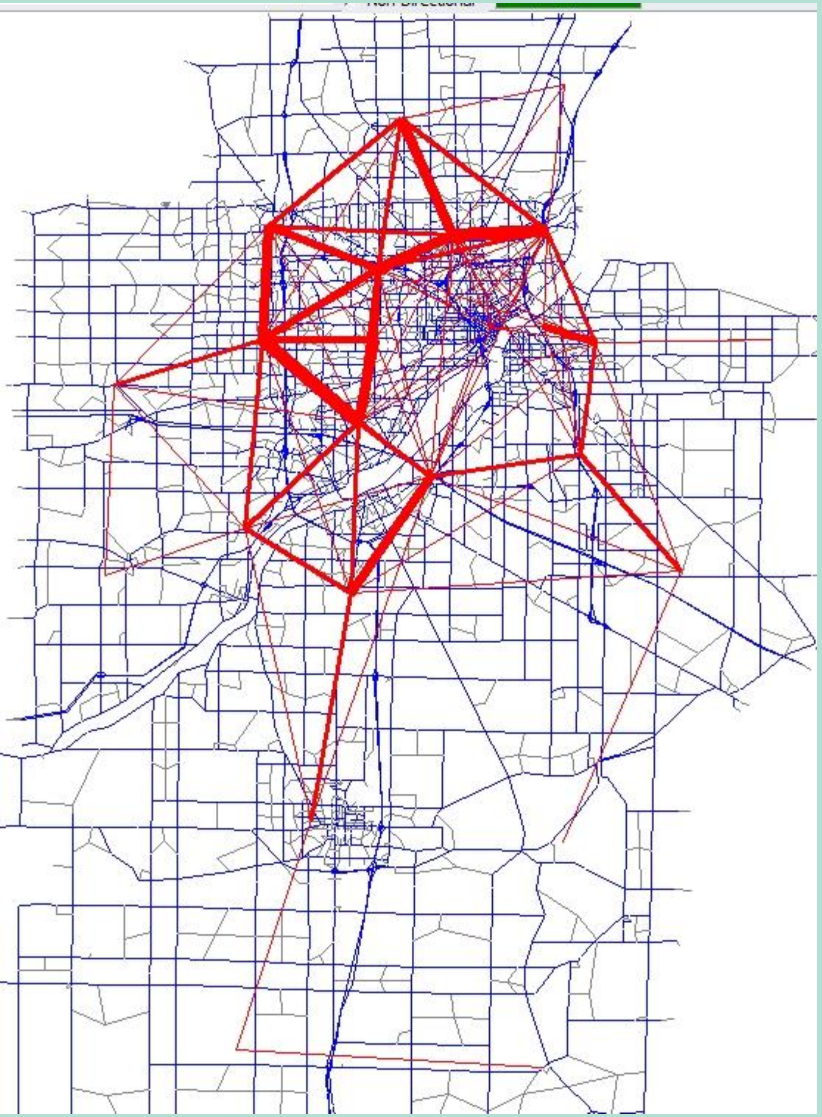
2016 LBS



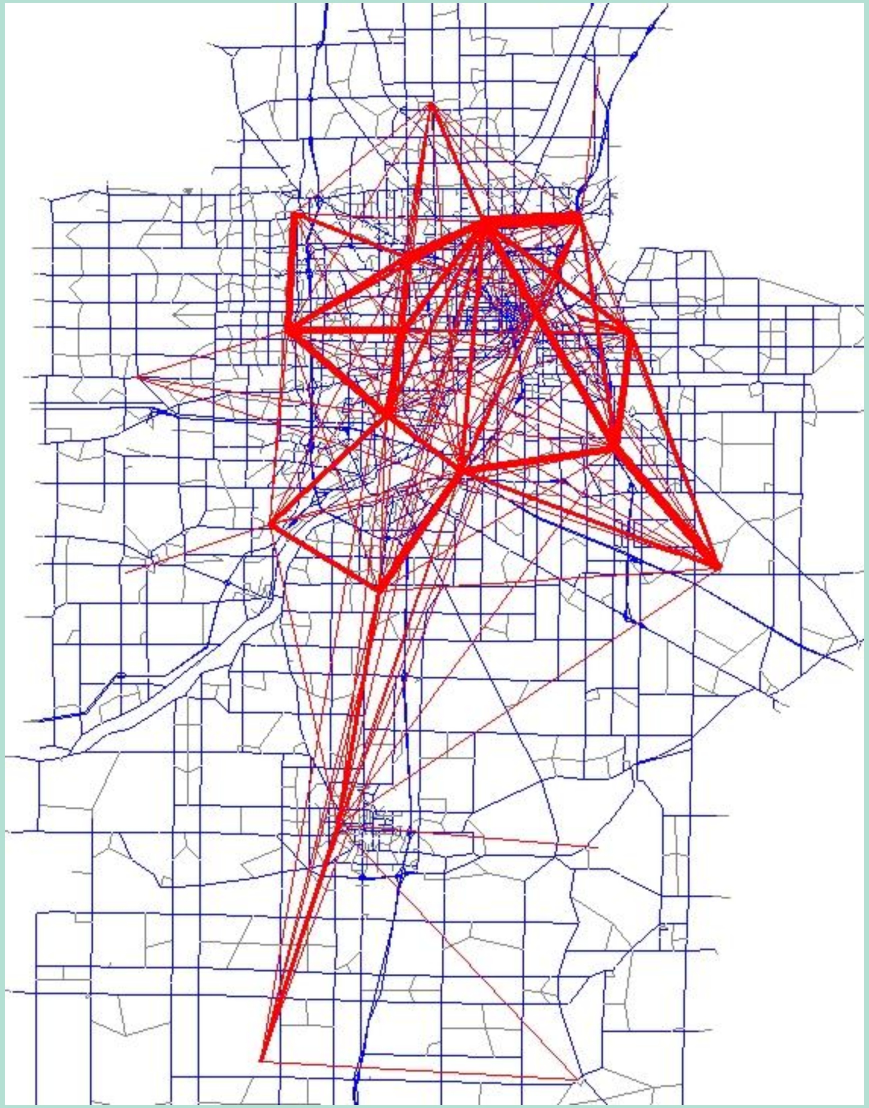
14-17 GPS

Desire Lines at District Level (Truck Trips)

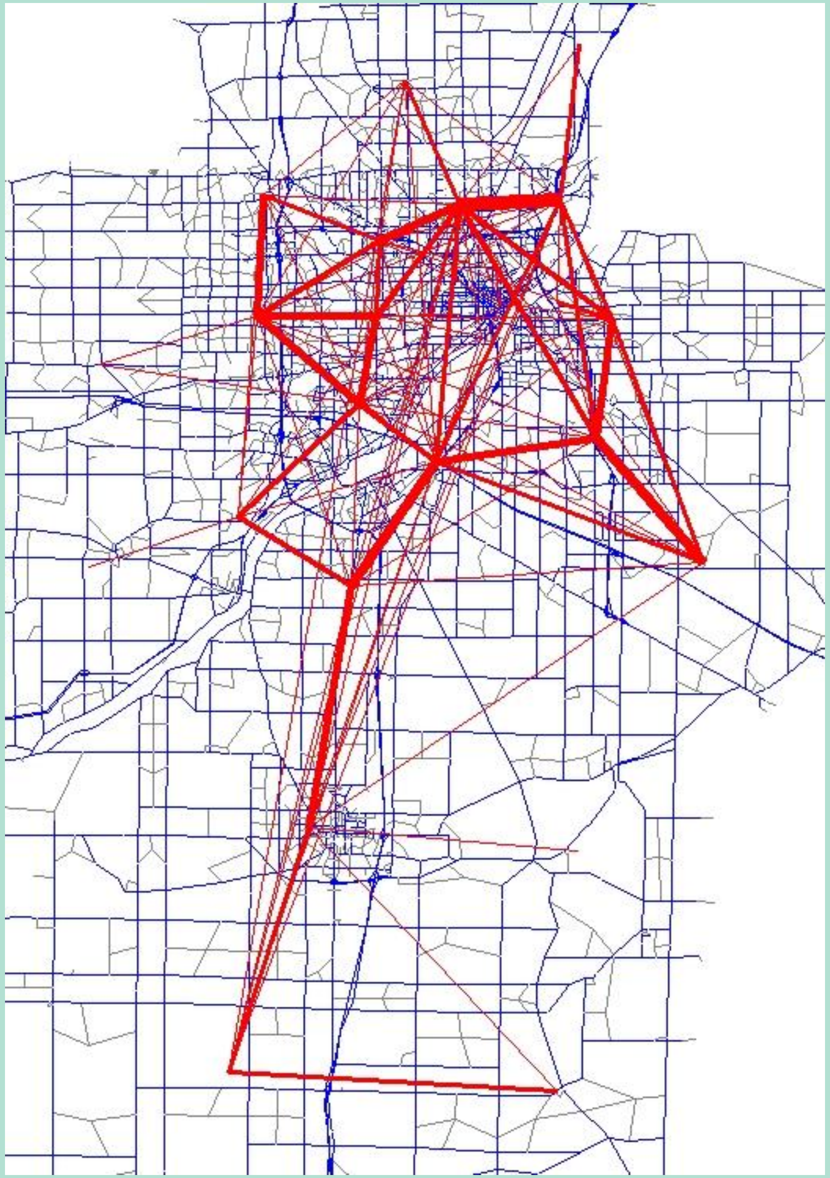
2015 Model



2016 GPS



14-17 GPS



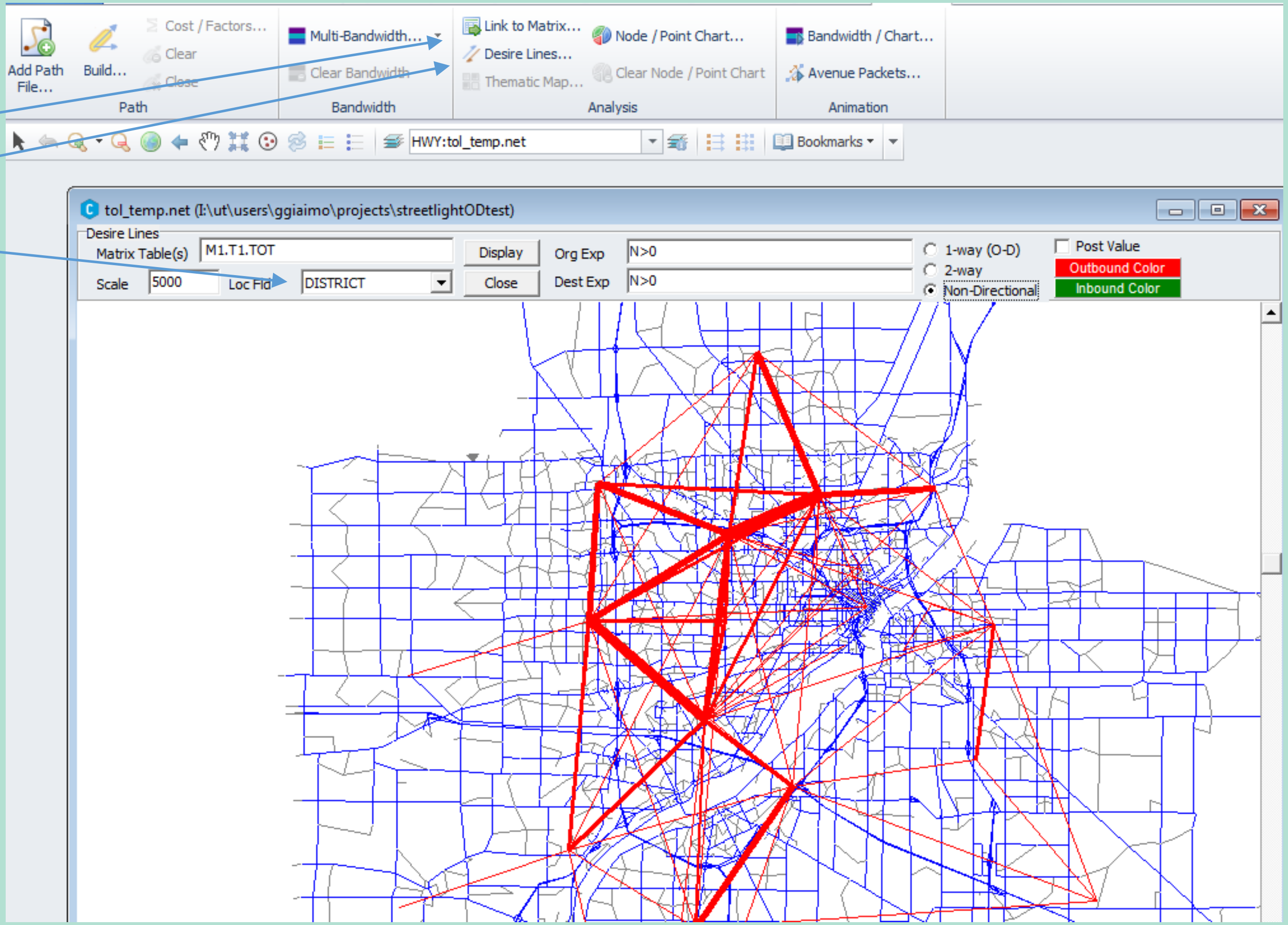
Desire Lines at District Level – Some Thoughts

- Compare better at District level as expected
- Even trucks look comparable
- Again the LBS looks like a better match to the model and at this level, where different I tend to believe the LBS data (based on completely unscientific local knowledge), so that indicates some utility of that data for model validation

Desire Lines at District Level – Some Thoughts

- Don't actually have a District GIS layer, found a feature where Cube will automatically aggregate TAZ trip tables to make District desire line plots if you have the District number on the network centroids

1.Open net and matrix
2. On the Analysis ribbon, link your matrix
3. Go to desire lines
4. Set this to your District field on the nodes



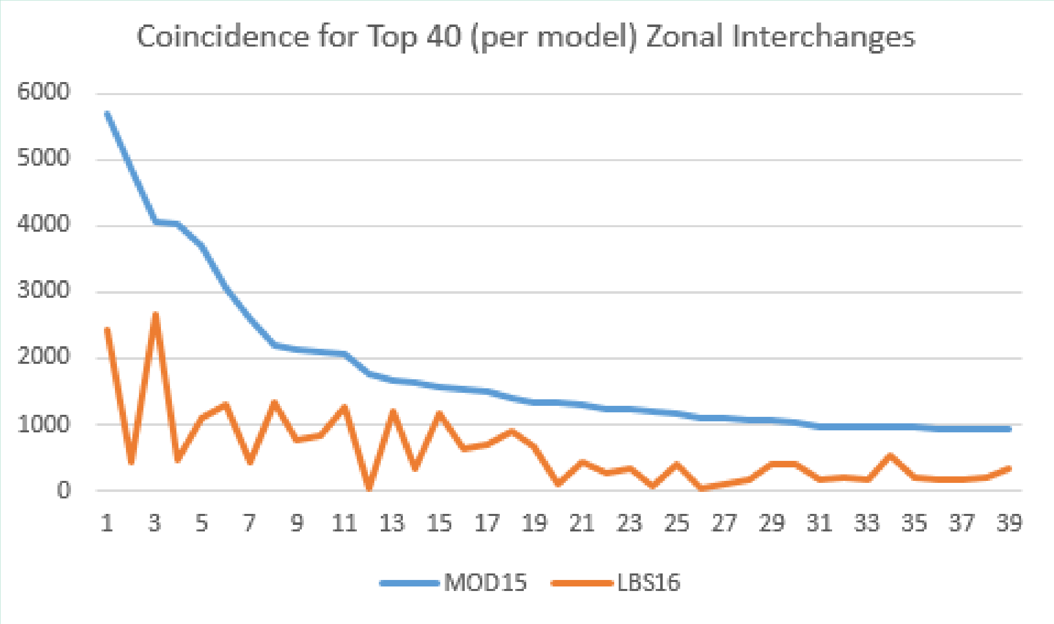
If you don't have the District on your network nodes it is easy to add with a little NETWORK script (note: I redefine District 0 as 99, Cube docs say a code of 0 means "don't plot", didn't test that)

```
RUN PGM=NETWORK PRNFILE="PRNFILE.prn" MSG="Add District to Nodes"
filei neti=tol15c.net
fileo neto=tol_temp.net

PHASE=NODEMERGE

DISTRICT=_NO.DISTRICT[1]   ;I'm just populating the district of the first outbound link
IF (DISTRICT=0)
DISTRICT=99
ENDIF

ENDPHASE

ENDRUN
```

Coincidence Ratios

- Compare how much of the matrices are the same

Explanation of Coincidence Ratio						
Trip Table 1 (sum 22)			Trip Table 2 (sum 22)			
3	8		2	6		
4	7		9	5		
Coincident Portion (min)			Non Coincident Portion (max-min)			
2	6		1	2		
4	5		5	2		
Ratio	0.63					
(sum the mins divided by the maxs)						

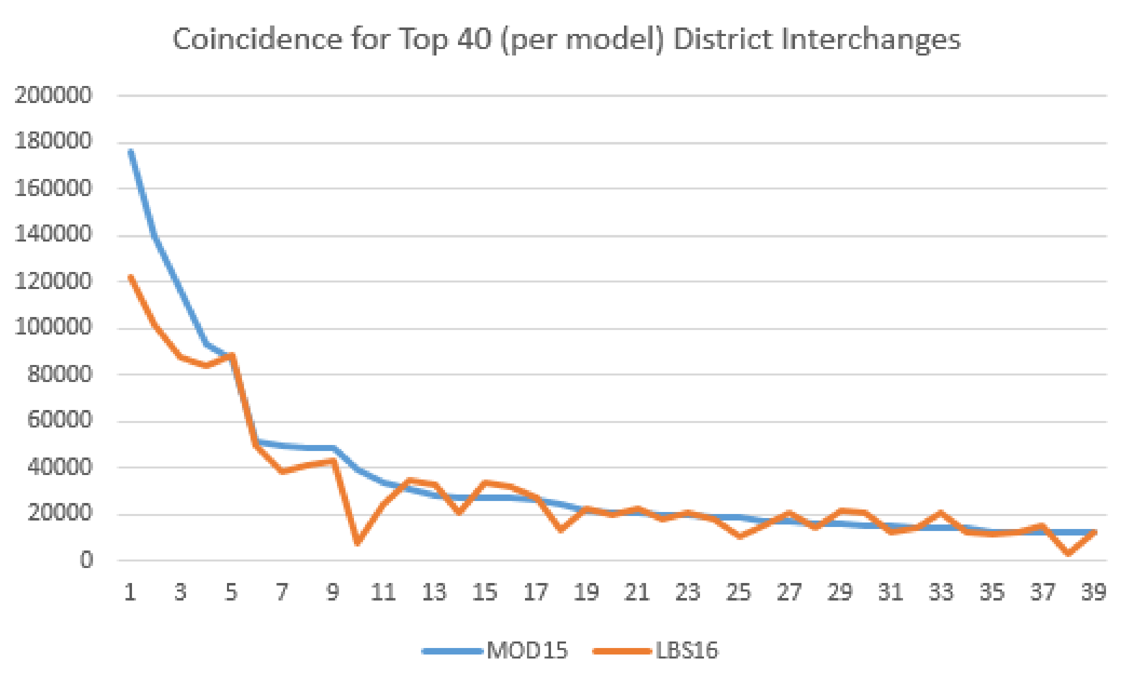


Trip Table TAZ Level Coincidence Ratios							
Test 1	2010 Mod	2010 Mod	2015 Mod	2015 Mod	2015 Mod	2016 GPS	2016 GPS
Test 2	2015 Mod	2045 Mod	2016 GPS	14-17 GPS	2016 LBS	14-17 GPS	2016 LBS
Car	0.96	0.85	0.28	0.29	na	0.63	na
Truck	0.97	0.87	0.24	0.24	na	0.63	na
Total	0.96	0.86	0.30	0.30	0.38	0.64	0.32

- Once again, LBS looks better (but again, can’t do trucks)
- GPS and LBS are no more similar to one another than they are to the model!

Coincidence Ratios

- Gets better at district level as expected
- But same conclusions as at TAZ level
- Looks like a lot of the trip length bias is gone at District level (I need to run some TLFD's but ran out of time)
- At this level LBS is more similar to the model than to the GPS!



Trip Table District Level Coincidence Ratios							
Test 1	2010 Mod	2010 Mod	2015 Mod	2015 Mod	2015 Mod	2016 GPS	2016 GPS
Test 2	2015 Mod	2045 Mod	2016 GPS	14-17 GPS	2016 LBS	14-17 GPS	2016 LBS
Car	0.98	0.92	0.61	0.61	na	0.89	na
Truck	0.99	0.95	0.59	0.61	na	0.87	na
Total	0.98	0.93	0.63	0.62	0.74	0.90	0.65

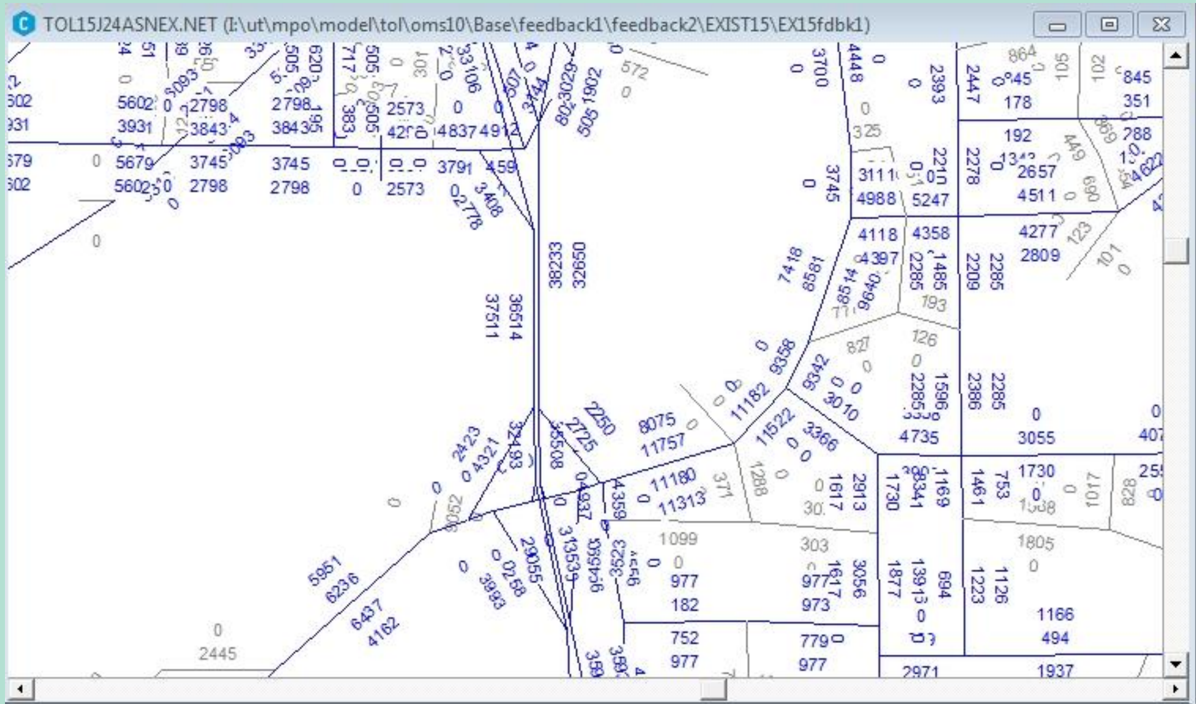
Note I actually had to go squeeze the matrices with a RENUMBER script to get to this, but the equivalency was easy to create since I could just use a DBF dump of the centroid nodes using that network node DISTRICT field I showed a few slides back

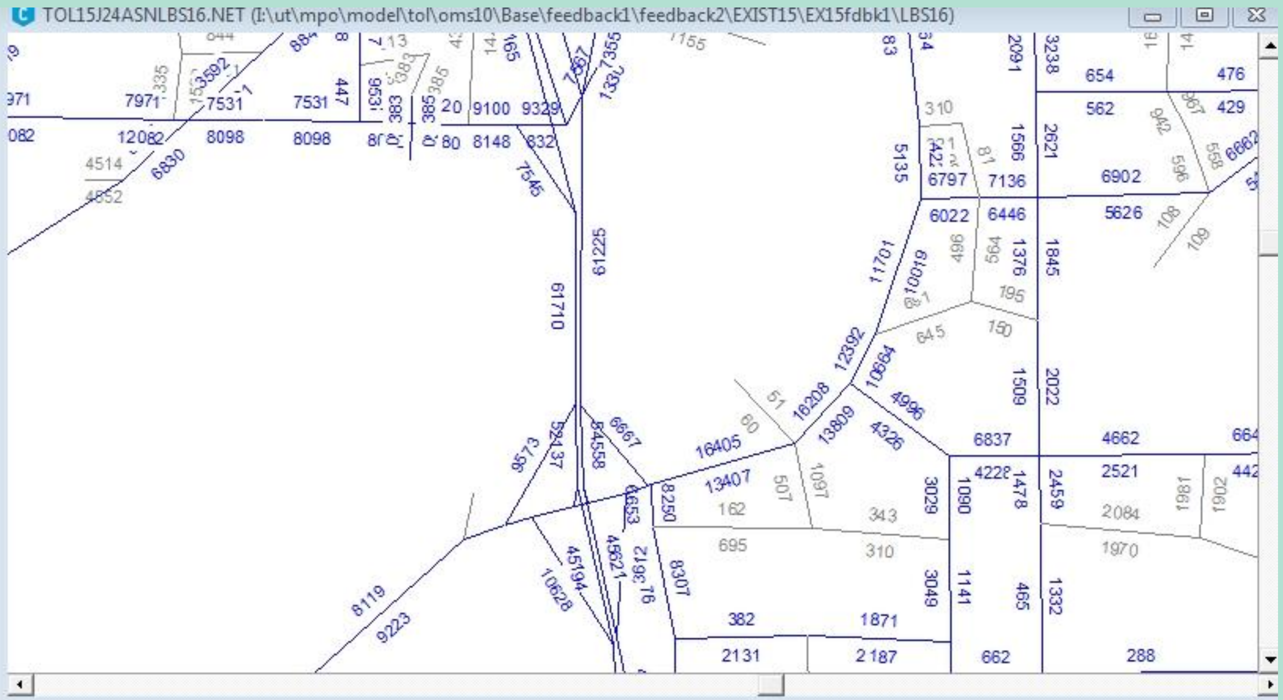
Assignments

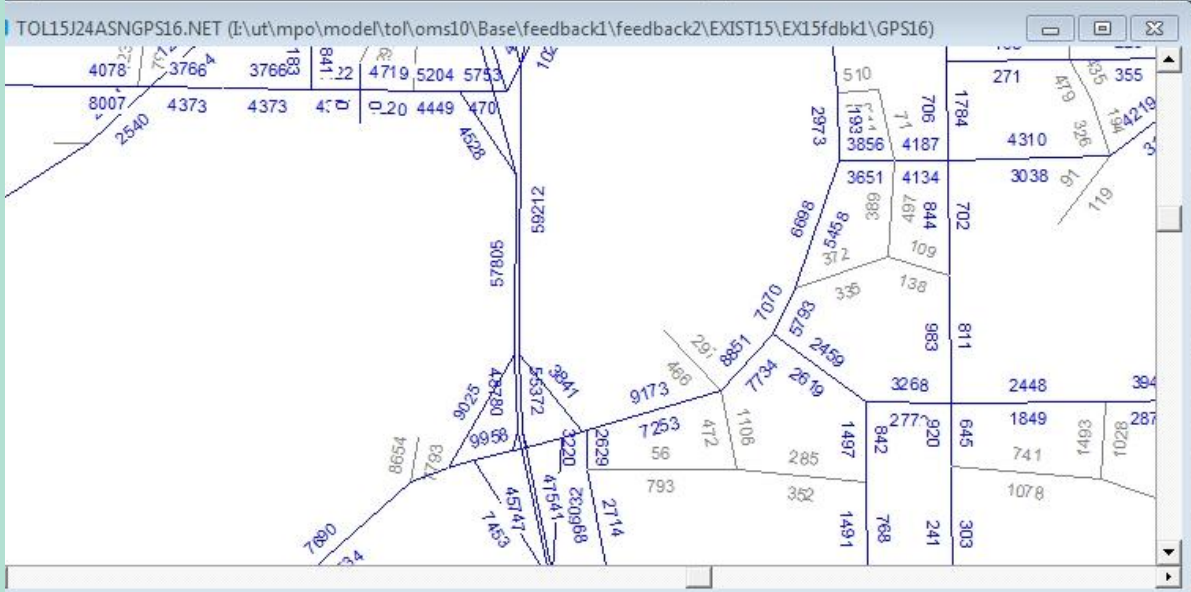
- Assigned the Streetlight trip tables to the network to compare to counts
- Used the IE/EE from model
- Since only downloaded daily trips, factored the model TOD results to match StreetLight daily totals (if StreetLight had trips where the model had none, applied a default TOD distribution to those trips, very rare since model trip table is mostly populated)
- Comparisons are with 2010 ADT
- Getting 15-24% more VMT due to the StreetLight long trip bias (recall the trip tables were scaled to the exact number of trips from the model)
- Spread across all functional classes and area types

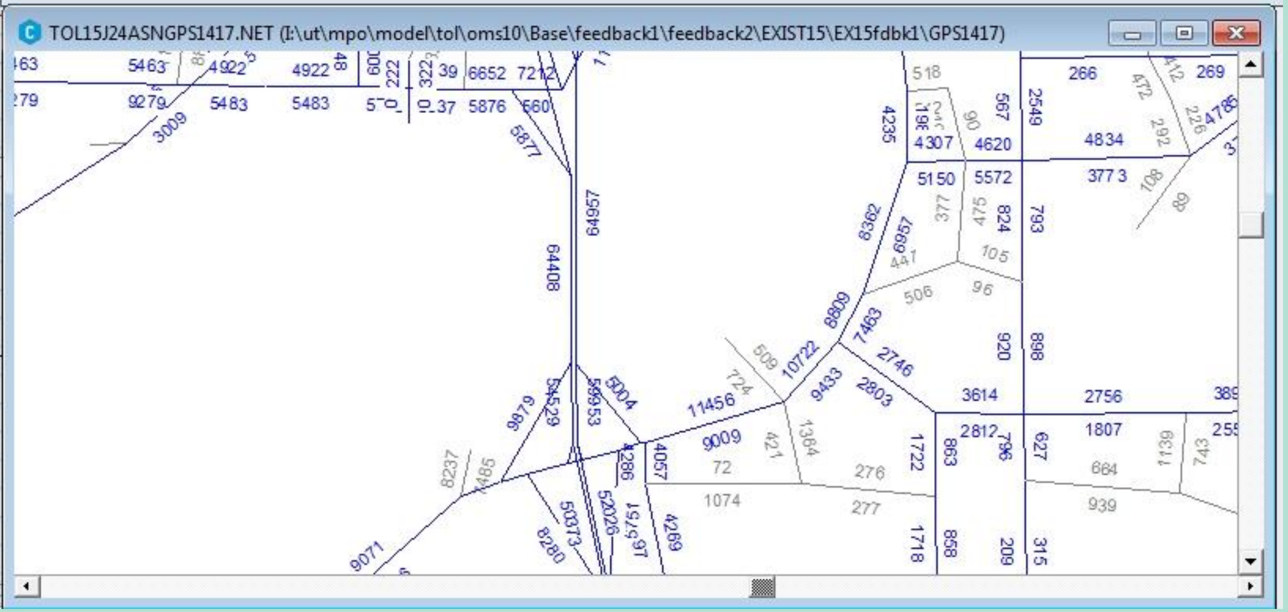
VMT Summary						
FACTYPE	Criteria	2010 Final	2015 Mode	GPS16	GPS1417	LBS16
Freeway	0.93-1.07	0.98	1.01	1.19	1.26	1.15
Expressway		0.90	1.00	1.08	1.13	1.06
Ramp		1.06	0.98	1.19	1.31	1.23
Arterial	0.90-1.10	1.01	1.01	1.14	1.22	1.25
Collector	0.85-1.15	0.98	0.97	1.13	1.23	1.23
Rural	0.85-1.15	0.97	0.97	1.13	1.24	1.30
AREATYPE						
Rural		0.97	0.99	1.02	1.10	1.07
Suburban		0.97	0.99	1.28	1.37	1.24
Urban		1.02	1.03	1.13	1.23	1.32
CBD		0.98	0.98	1.20	1.31	1.28
Outlying BD		0.98	0.96	1.13	1.17	1.14
Total		0.98	1.00	1.15	1.24	1.20

Assignments







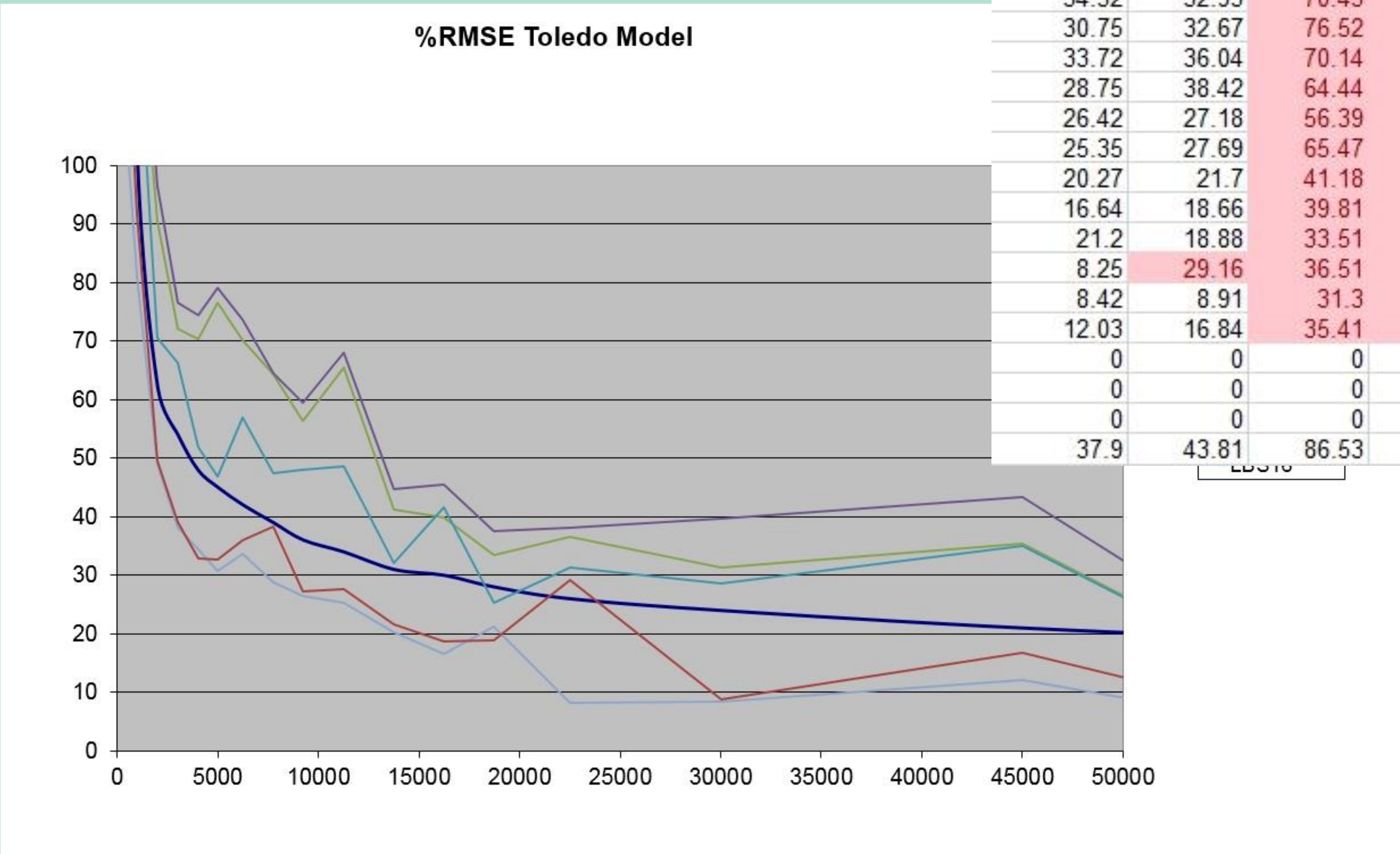


Assignments

- Of course, the model was calibrated to counts, including the network used for these assignments

%RMSE Total				
2010 final	2015 mode	GPS16	GPS1417	LBS16
116.36	140.79	200.3	224.02	189.85
81.15	91.86	132.44	141.48	127.16
48.98	49.59	90.63	96.63	70.48
38.23	39.05	72.02	76.61	66.26
34.52	32.93	70.45	74.38	51.97
30.75	32.67	76.52	79.04	46.86
33.72	36.04	70.14	73.71	56.88
28.75	38.42	64.44	64.5	47.43
26.42	27.18	56.39	59.51	47.97
25.35	27.69	65.47	67.98	48.57
20.27	21.7	41.18	44.66	32.13
16.64	18.66	39.81	45.49	41.57
21.2	18.88	33.51	37.53	25.26
8.25	29.16	36.51	38.06	31.29
8.42	8.91	31.3	39.73	28.67
12.03	16.84	35.41	43.32	35.11
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
37.9	43.81	86.53	94.87	72.46

%RMSE Truck				
2010 final	2015 mode	GPS16	GPS1417	LBS16
156.62	173.94	248.71	234.34	260.58
67.73	74.65	98.86	94.15	88.32
58.13	59.51	69.22	65.51	56.54
53.2	59.15	67.63	67.39	54.04
46.97	65.03	69.78	71.15	60.18
18.41	34.01	54.51	47.08	50.51
13.06	40.69	68.44	62.93	46.38
17.83	12.47	46.85	41.67	31.83
29.47	29.27	34.97	33.44	31.31
17.53	22.52	49.11	45.42	35.79
13.91	31.38	43.17	39.35	40.44
5.48	6.31	21.45	16.26	20.7
6.77	60.27	61.28	61.2	60.21
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
74.23	101.60	143.40	135.13	122.44



Assignments

- And just jumping to 2015 adds a fair amount of error itself
- LBS looks better than GPS again

%RMSE Total				
2010 final	2015 mode	GPS16	GPS1417	LBS16
116.36	140.79	200.3	224.02	189.85
81.15	91.86	132.44	141.48	127.16
48.98	49.59	90.63	96.63	70.48
38.23	39.05	72.02	76.61	66.26
34.52	32.93	70.45	74.38	51.97
30.75	32.67	76.52	79.04	46.86
33.72	36.04	70.14	73.71	56.88
28.75	38.42	64.44	64.5	47.43
26.42	27.18	56.39	59.51	47.97
25.35	27.69	65.47	67.98	48.57
20.27	21.7	41.18	44.66	32.13
16.64	18.66	39.81	45.49	41.57
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8.25	29.16	36.51	38.06	31.29
8.42	8.91	31.3	39.73	28.67
12.03	16.84	35.41	43.32	35.11
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
37.9	43.81	86.53	94.87	72.46

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156.62	173.94	248.71	234.34	260.58
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58.13	59.51	69.22	65.51	56.54
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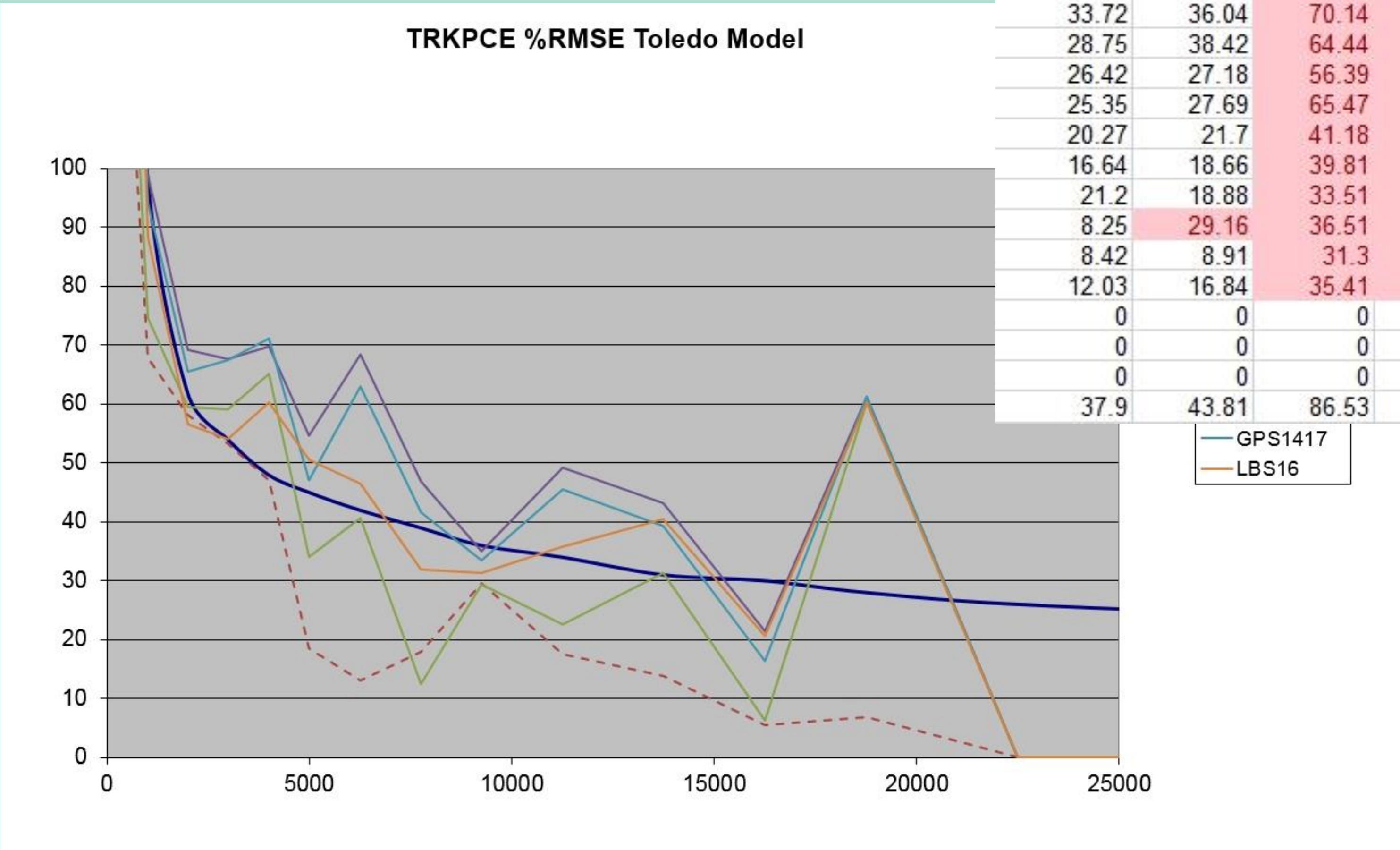
18.41	34.01	54.51	47.08	50.51
13.06	40.69	68.44	62.93	46.38
17.83	12.47	46.85	41.67	31.83
29.47	29.27	34.97	33.44	31.31
17.53	22.52	49.11	45.42	35.79
13.91	31.38	43.17	39.35	40.44
5.48	6.31	21.45	16.26	20.7
6.77	60.27	61.28	61.2	60.21
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
74.23	101.60	143.40	135.13	122.44



Conclusions and Next Steps

- LBS data at District level looks promising for validation of car trips
- GPS data has already been shown to be useful for EE trips and may be reasonable for IE and truck trip modeling at TAZ level (partly because we have nothing better) but needs more study
- Trip length bias in the purchased data is a serious concern
- Need to look more into income bias as well if using the GPS data
- Haven’t looked at the TOD specific data yet, however, the finer you chop it the worse it gets
- I think (but need to investigate further) it makes most sense to use only spring/fall data to assemble the trip tables, didn’t see huge difference in this analysis
- With LBS can also get pseudo-work/non-work breakdowns, need to compare those to model as well as CTPP JTW
- Need to test on other areas
- Need to decide if some other StreetLight factoring scheme is in order to try to ameliorate trip length bias, probably need to look more directly at the TLFD’s as part of that
- Need to decide how to use in validation and what criteria
- All my scripts are ad hoc, once I dress them up a little I’ll make them available